

Can you share an example of how precision robots are applied in real-world manufacturing?

A strong example is an electric vehicle battery assembly. EV packs involve handling large numbers of individual battery cells, each requiring high-speed movement combined with extremely precise positioning. Processes such as cell placement, welding, and adhesive dispensing are highly critical, as even minor deviations can affect safety, reliability, and overall performance.

Our robots execute these operations with tight motion control and high repeatability, enabling fast throughput while maintaining the precision required for battery manufacturing.

How do sensor and drive advances boost compact robot accuracy, and how has Epson used them?

Advances in sensing and drive technologies now enable higher accuracy without increasing robot size. We achieve this through Gyro Plus technology, which allows robots to remain compact while operating at higher speeds and precision. The system continuously captures motion feedback during operation and rapidly adjusts control in real time. This closed-loop feedback provides faster response, improved stability, and higher positioning accuracy, enabling compact robots to deliver performance levels previously achievable only in larger systems.

How critical is control architecture for advanced robotics, and how is Epson addressing it in its platforms?

Control architecture is becoming as important as robot hardware, especially with the shift towards remote operation and continuous monitoring. We help manufacturers capture real-time data from robots and associated automation via RC+ and remote monitoring systems, enabling supervision, diagnostics, and control without on-site presence. With growing demand for cloud-enabled, data-driven automation, we continue to advance RC+ and related

services to support next-generation manufacturing.

Is robotics increasingly being defined by software, AI, and control intelligence rather than hardware?

At the factory level, the physical robot will always be essential, but the primary differentiation is increasingly shifting to software and control intelligence. Robot motion and process logic can now be developed, optimised, and supervised remotely through software platforms. AI-based control enables repetitive operations to run autonomously, reducing the need for continuous human intervention, including routine monitoring. This evolution extends beyond basic motion control to full application-level intelligence, where software and artificial intelligence enhance efficiency, adaptability, and operational decision-making.

How is the convergence of vision systems and robotics changing automation, and how do you implement it?

The latest robots, especially in pick-and-place applications, use machine vision to recognise part shapes, colours, and positions. This ensures precise handling, while force sensors regulate the correct pressure for tasks such as screw tightening or component placement. The system integrates the robot arm, vision cameras, force sensors, feeder units, and RC+ software for smooth operation. Feeder units align incoming components through vibration and nano-motion so the vision system can verify orientation before robotic handling. Together, these elements create a fully automated solution capable of high precision in complex factory tasks.

How does edge AI enhance real-time decision making, motion reliability, and productivity in robotics?

Edge AI is increasingly integrated into robotics to handle repetitive tasks and reduce reliance on human intervention. It allows robots to

react instantly, for example, restarting operations automatically without a manual reset. Combined with technologies such as Zero Plus and SafeSense, AI improves precision and reliability in complex operations. It also collects data from multiple components across the automation system, enabling real-time insights that enhance productivity and overall efficiency.

Can you share an example of how AI improves robot vision performance?

Without AI, identifying or rejecting parts is slower and more susceptible to variations in lighting and environmental conditions. With AI, we can calibrate lighting automatically and optimise image processing in real time, resulting in greater accuracy and consistency. Although gains vary by application, we typically see lower rejection rates, faster throughput, and higher overall productivity compared with conventional vision systems.

What non-technical barriers limit robotics adoption in India, and how do you address them?

When we engage with MSME customers, we see two main barriers. First, **there is doubt about whether automation can be implemented successfully at all.** Many customers have limited knowledge and are unsure if their processes can be automated. To address this, we use 3D simulators to demonstrate the feasibility of automation before actual deployment. Second, there are concerns about return on investment. Customers need clarity on how automation improves productivity, efficiency, and consistency compared to manual operations, where errors can reduce output. **We provide detailed RoI calculations showing 24/7 operation, higher efficiency, and reduced human error.** Addressing both feasibility and RoI concerns helps overcome the primary adoption barriers for MSMEs. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/Intmar03>

